GEORGE THOMAS

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SKILLS & INTERESTS

- **Technical Skills:** Verilog, Cadence Xcelium, Cadence Genus, Cadence Virtuoso, Arduino, Python, C, KiCAD
- **Soft Skills:** Team Collaboration, Adaptability, Organisational Skills, Conflict Resolution, Communication
- **Interests:** RTL Design, FPGA Design Flow, ASIC Design Flow, Robotics and Automation, CAD Design, STA, Graphic Design, PCB Design

EDUCATION

- **Govt. Model Engineering College** — **8.93 |** 2027
  KTU, B.Tech in Electronics Engineering in VLSI Design and Technology
- **Aramaia International Residential School** — **98.5%** | 2022
  State, 12^{th}
- **Msgr. Raymond Memorial School** — **91.2%** | 2020
  CBSE, 10^{th}

WORK EXPERIENCE

- **Semi-Conductor Laboratory (SCL), Govt. of India** — *1 Month*
  Project Trainee
  Technologies Used*: LPCVD (Low-Pressure Chemical Vapor Deposition), LTO (Low-Temperature Oxide), Silicon Doping, Wafer Fabrication*
  Gained practical experience in semiconductor manufacturing under a Ministry of Electronics & Information Technology (MeitY) industrial internship, with direct exposure to front-end fabrication techniques in 6” MEMS and Diffusion Doping LPCVD LTO Oxide.
- **CUSAT Department of Electronics** — *2 Weeks*
  Research Intern
  Technologies Used: *Cadence Xcelium, Cadence Genus, Xilinx Vivado*
  Designed and evaluated 5 comparator architectures and implemented 32-bit Matrix Multiplication, 1D Convolution and Dot Product using Cadence Xcelium, gaining hands-on experience in VLSI digital circuit design, simulation and performance evaluation.

PROJECTS

- **LiDAR Voxelization / Spatial Binning IP** — **Team Size: 4**
  RTL Design Engineer — *Ongoing*
  Technologies Used: *Cadence Xcelium, Cadence Genus, Xilinx Vivado*
  Developing a fixed-point spatial voxelization algorithm within a 50m x 40m hardware Region of Interest (ROI), proving 97% data sparsity to justify a memory-efficient hash table hardware architecture.
- **8-Bit Cryptographic Processor (TinyCrypto-8)** — **Team Size: 3**
  RTL & Architecture Designer — *Ongoing*
  Technologies Used: *Verilog, Vivado, OpenLane (OpenROAD), Tiny Tapeout*
  Architected a custom 8-bit Verilog microprocessor featuring a 2-stage pipeline, 16-instruction ISA, and an integrated LFSR crypto-engine for real-time encryption. Optimized with zero-latency instruction fetching, the design is currently slated for 130nm physical manufacturing via the Tiny Tapeout ASIC program.
- **RescueLink: LoRaWAN Emergency Response System** — **Team Size: 12**
  Hardware Integration Engineer — *5 Months*
  Technologies Used: *Arduino MKR WAN 1310, LoRaWAN Gateway, Embedded C*
  Designed and developed RescueLink, an affordable SOS system under IEEE SPS Kerala Chapter and ICFOSS using Arduino MKR WAN 1310 and LoRaWAN to enable long-range emergency communication in regions with 0% cellular coverage.

COURSES AND CERTIFICATIONS

- Pursuing an **Honours Degree** in **Electronics Design and Automation** from **KTU**.
- Strengthened practical skills in **Hardware Modelling using Verilog** via hands-on training offered by **IIT Kharagpur**.
- Received a certification in **VLSI for Beginners** through **C2S**, offered in association with **NIELIT**.

KEY POSITIONS

- **Training Cell volunteer**, Govt. Model Engineering College.
- Served as the **Kochi Hub Student Representative** at **IEEE SPS Kerala Chapter 2025**, the technical society chapter, contributing to the organisation and coordination of regional technical events across the Kochi Hub.
- Managed leadership responsibilities at **IEEE CASS Kerala Chapter 2025** in the position of **Event Head**, managing and organising events.
- Acted as the **Design Head** of **CogniCor AICTE IDEALab 2025**, the fabrication and ideation lab, overseeing the promotion of technical and project works.
- Undertook the role of **Chair** at **IEEE SPS MEC SB 2026**, responsible for directing the strategic planning and execution of workshops.
- Worked as the **Operations Manager** at **IEEE MEC SB 2025**.
- Fulfilled duties of **Design Head** at **Alumni Relation Cell 2025**, coordinating publicity materials and branding for major events, including the graduation ceremony.

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ACHIEVEMENTS AND ACTIVITIES

- Secured **1st prize** in **Untangle 2023**, a circuit debugging competition held during **Takshak'23**, conducted by **Mar Athanasius College of Engineering**, **Kothamangalam**.
- **2nd Prize, Open Silicon Microelectronics Bootcamp**: Awarded for the end-to-end RTL design, synthesis, and physical tapeout of an 8-bit pipelined microprocessor via **Tiny Tapeout**.
- Selected for the **Millennium Fellowship 2024** to develop a project advancing the **United Nations Sustainable Development Goals**.
- Gained hands-on exposure to **SOS Systems** through a **Project Building Program**, conducted by **IEEE SPS Kerala Chapter**.
- Participated in **Hands-on Workshop on VLSI & FPGA Design**, organised by **IIT Palakkad**.
- Volunteer of **ICTEST'24**, **ICTEST'25** and **ICTEST'26**, the international conference of **Model Engineering College** to bring researchers and students together.
- **Hobbies:** Badminton, Graphic Designing, Hiking

REFERENCES

- **Prof. Dr. Jayachandran E S**, Principal, Govt. Model Engineering College, Kochi, Email ID: **principal@mec.ac.in**
- **Prof. Dr. Jobymol Jacob**, HOD, Electronics and Communication Engineering, Govt. Model Engineering College, Kochi, Email ID: **hodec@mec.ac.in**